

ISAC-EMPOWERED LEO CONSTELLATION NETWORKED SENSING: CHALLENGES AND OPPORTUNITIES

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ABSTRACT

The pursuit of seamless global coverage in sixth-generation (6G) mobile networks has driven the development of low earth orbit (LEO) constellation networks. In addition to communication, the emerging 6G applications, such as low-altitude economy and smart city, also require high-accuracy, large-scale, and continuous sensing. Integrated sensing and communication (ISAC) has emerged as a key enabler, which could empower LEO constellations with dual capabilities of communication and sensing. With the implementation of networked sensing in ISAC-empowered LEO constellations, the requirements of emerging 6G applications can be satisfied. This article presents a comprehensive overview of networked sensing in ISAC-empowered LEO constellations. We first introduce the concepts, sensing types, application scenarios, and payload enhancements. Then, we delve into critical technical challenges of networked sensing in ISAC-empowered LEO constellations and discuss the promising solutions tailored to the unique characteristics of LEO constellations, such as high mobility and long distances to the targets. Finally, we highlight future research trends to further promote the performance of ISAC-empowered LEO constellations.

I. INTRODUCTION

Low earth orbit (LEO) constellation networks offer seamless global communication connectivity [1]. Recognizing this potential, the 3rd generation partnership project (3GPP) initiated the standardization of non-terrestrial networks (NTNs), beginning with foundational work in Release 15 [2]. This effort has rapidly evolved through subsequent Releases 16-19, enhancing functionalities and expanding support to higher frequency bands [3]. This ongoing standardization lays a solid foundation for future sixth-generation (6G) mobile networks, envisioning LEO satellites as a cornerstone of 6G infrastructure for ubiquitous connectivity [4]. However, in addition to

wide-area and reliable communication, emerging 6G applications, such as the low-altitude economy (LAE) and smart city, require high-accuracy, large-scale, and continuous sensing, which can be realized by cooperative sensing among multiple nodes. As a result, there is a growing requirement to integrate sensing functionalities into LEO constellation networks to realize the dual functions of communication and sensing, supporting the emerging 6G applications.

Advances in integrated sensing and communication (ISAC) offer a promising path to meet these dual requirements [5], [6]. The ongoing evolution of NTN standards towards higher frequency bands is a critical catalyst for this integration, as the wider bandwidths unlock the potential for high-resolution sensing. Consequently, ISAC-enabled LEO constellation networks are now attracting significant attention from both academia and industry to provide both seamless communication and high-accuracy sensing [4]. This concept should be distinguished from traditional satellite remote sensing, which focuses on static imaging with a lower temporal resolution. In contrast, ISAC-empowered LEO sensing provides the high-refresh-rate tracking of dynamic targets that is essential for emerging 6G applications. As summarized in Table I, compared with traditional ISAC-enabled terrestrial and aerial networks, ISAC-empowered LEO constellations exhibit significant advantages, including stronger space-to-ground line-of-sight (LoS) links, global coverage, and the ability to support large-scale multi-node cooperation.

However, LEO satellites have unique characteristics, such as high altitudes (500 ~ 2000 km) and extreme orbital velocities (6.9 ~ 7.61 km/s) [2]. These factors pose several critical challenges to achieving efficient and robust networked sensing.

- **ISAC waveform design for LEO constellations:** Conventional orthogonal frequency division multiplexing (OFDM) waveforms struggle to satisfy the stringent unambiguous sensing requirements in LEO constellations, where relative distance and velocity exceed 500 km and 6.9 km/s, respectively. Moreover, waveforms

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Sensing Performance Metrics	ISAC-Enabled Terrestrial Networks	ISAC-Enabled Aerial Networks	ISAC-Empowered LEO Constellations
Coverage	Local area (single base station (BS) coverage radius < 10 km)	Medium area (unmanned aerial vehicle (UAV) swarm coverage radius of 50 – 100 km)	Ultra-wide area (inter-constellation cooperation achieving global coverage)
Dynamic adaptive requirement	Medium (cellular handover can rely on the terrestrial core network)	High (UAV swarm require dynamic formation)	Ultra-high (performing topology transformation while moving at high speed)
Sensing detection probability	High (multi-BS cooperative detection probability higher than single-BS)	Medium-High (flexible perspective, but susceptible to environmental interference)	Medium-high (flexible perspective, but susceptible to synchronization accuracy and channel effects)
Sensing resolution	Medium-high (customizable radar waveforms, but limited by occlusion)	Medium (diverse perspectives, but requires flexible waveform design)	Medium (diverse perspectives, but requires flexible waveform design and gain compensation)
Sensing continuity	Medium (multi-BS cooperative sensing)	High (UAV swarm cooperative sensing)	Ultra-high (Inter-constellation/intra-constellation cooperative sensing)
Multi-node cooperation capability	Medium-high (multiple BSs, BS-to-terminal)	High (unique UAV swarm cooperation)	Ultra-high (constellation-level distributed and integrated space-air-ground cooperation)
Latency sensitivity	Low (terrestrial fiber backhaul < 5 ms)	Medium (space-to-ground link latency of 10 – 30 ms)	High (inter-satellite/ground-satellite latency accumulation, requiring on-board preprocessing)
Synchronization accuracy requirement	Medium (cooperative sensing requires ~100 ns)	High (dynamic beam alignment requires ~10 ns)	Ultra-high (multi-satellite fusion requires < 10 ns [2])
System computational capacity	High (BSs connected to cloud/edge servers)	Medium-low (UAV computational capacity constrained)	Low (requires distributed lightweight algorithms)
Interference resistance	Medium (multi-user interference controllable, but limited by dense deployment)	Low (complex interference in low-altitude environments)	Medium (frequency band isolation, but challenged by inter-satellite interference)
System complexity	High (joint scheduling algorithms)	Medium-high (dynamic network topology)	Ultra-high (cross-layer optimization and distributed coordination)

TABLE I. Sensing performance metrics comparison of the ISAC-enabled terrestrial networks, ISAC-enabled aerial networks, and ISAC-empowered LEO constellations [2]. Note: The terms used for performance levels are ranked qualitatively as follows: Ultra-high > High > Medium-high > Medium > Medium-low > Low.

tailored for high-accuracy sensing typically exhibit high peak-to-average power ratio (PAPR), which conflicts with the low-PAPR constraints of power-limited LEO satellite.

- **Target sensing in LEO constellations:** The limited transmit power of LEO satellite and large propagation loss (e.g., ionospheric attenuation and atmospheric absorption) lead to extremely low signal-to-noise ratio (SNR) of received echo signal, which severely hampers stable and reliable sensing performance, particularly for the target with low radar cross-sections (RCS).
- **Continuous target tracking in LEO constellations:** The high orbital velocity and dynamic topology of LEO satellite network result in rapidly varying and nonlinear target prediction models. These characteristics pose challenges to maintaining stringent tracking performance, specifically, exceeding 99.9% continuity and 2 Hz update rate.

To address the aforementioned challenges, several technical and survey studies have been conducted. In terms of technical studies, Egea-Roca et al. [7] proposed a chirp spread spectrum-based ISAC waveform specifically designed for LEO satellite positioning. By dynamically tuning the chirp period and constellation parameters, their approach achieves an effective trade-off between sensing accuracy and communication capacity, thereby enhancing the overall system robustness. Wang et al. [4] proposed a multi-satellite cooperative ISAC framework that enhances both data transmission rate and sensing

accuracy through the joint optimization of communication beams and sensing signals. However, the existing studies on networked sensing in ISAC-empowered LEO constellations primarily focus on performance optimization, while lacking a comprehensive study on the networked sensing of ISAC-empowered LEO constellations.

In terms of survey studies, You et al. [8] reviewed the concepts and challenges associated with ISAC-enabled massive multiple-input multiple-output (MIMO) LEO satellite networks, including payload constraints, traffic distribution, and mobility management. Gu et al. [9] provided a summary of key enabling technologies and future research trends for ISAC in integrated satellite-terrestrial networks, covering the aspects of interference cancellation, distributed cooperative sensing, and frame structure design.

While these studies offer valuable insights, few of them have focused on the specific sensing types, payload enhancements, waveform design, and advanced signal processing that are required for target detection and tracking within networked ISAC-empowered LEO constellations. Hence, this article presents a comprehensive state-of-the-art overview of networked sensing in ISAC-empowered LEO constellations, with a particular focus on the key challenges and solutions. The main contributions of this article are summarized as follows.

- **Networked sensing in ISAC-empowered LEO constellations:** We first introduce the core concepts, sensing types, representative use cases, and requirements of networked sensing in

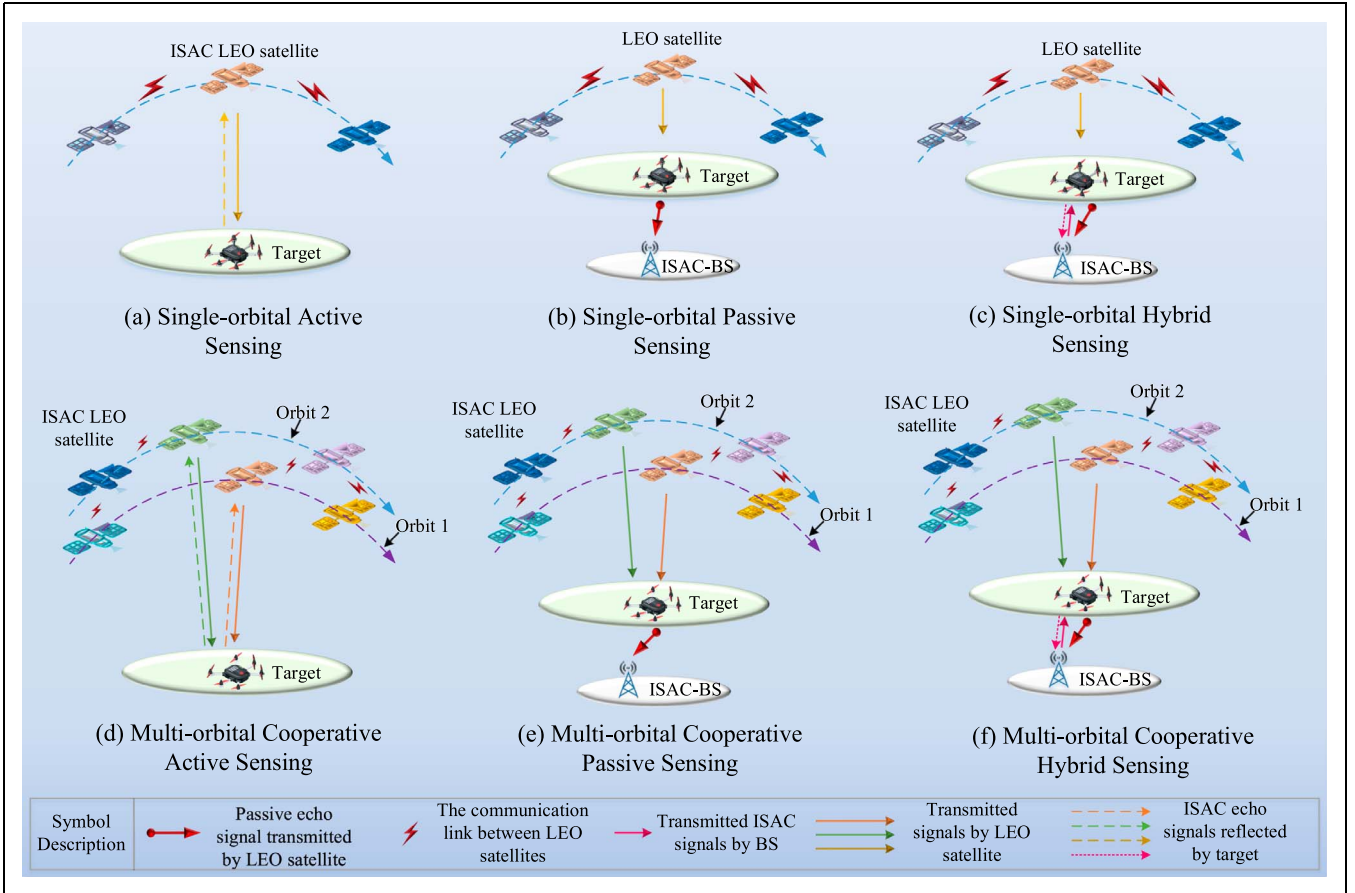


FIG. 1. Taxonomy of networked sensing types in ISAC-empowered LEO constellations.

ISAC-enabled LEO constellations. We then discuss the principal technical challenges and potential solutions in terms of waveform design, target sensing and tracking, followed by future research trends.

- **ISAC waveform design:** We analyze the unique challenges of waveform design in LEO constellations, including the requirement for extended unambiguous range and velocity, and the trade-offs among peak-to-average power ratio (PAPR), communication, and sensing. Then, new ISAC waveform design schemes are proposed, and their effectiveness is validated through simulation.
- **Target sensing and tracking:** We investigate the core challenges in target sensing and tracking of ISAC-empowered LEO constellations, including synchronization, sensing information fusion, optimal satellite node selection, and target tracking algorithm design. In particular, we propose a multi-satellite sensing information fusion framework based on convolutional neural networks (CNNs), and demonstrate its superior performance over state-of-the-art baselines via simulation results.

II. FOUNDATION OF ISAC-EMPOWERED LEO CONSTELLATION NETWORKED SENSING

This section introduces the types of networked sensing, use cases, and requirements of networked sensing in ISAC-empowered LEO constellations. Then, we explore the feasibility of implementation and key technical challenges of networked sensing in ISAC-empowered LEO constellations.

A. TYPES OF NETWORKED SENSING

Due to the continuous orbital mobility of LEO satellites, their coverage footprints change rapidly over time. As a result, maintaining continuous sensing coverage over a target region requires frequent satellite handovers [4], which is a basic operation when realizing networked sensing in LEO constellations.

Without loss of generality, assuming a UAV as the target and a ground ISAC base station (BS) as the receiver, the types of networked sensing can be classified into *active sensing*, *passive sensing*, and *hybrid sensing*, each of which could operate in either a *single-orbital* or *multi-orbital cooperative sensing*. The taxonomy is illustrated in Fig. 1 and described as follows.

- **Active, passive, and hybrid sensing:** *Active sensing* applies the echo signals of the signals transmitted by ISAC LEO satellite for sensing (e.g., Fig. 1(a)). *Passive sensing* occurs when an ISAC BS utilizes reflected downlink communication signals from the LEO satellite for sensing (e.g., Fig. 1(b)). *Hybrid sensing* combines active and passive sensing, where the ISAC BS fuses the echo signals of its own transmit signals and the reflected downlink signals of LEO satellites to enhance sensing accuracy.
- **Single-orbital and multi-orbital cooperative sensing:** *Single-orbital sensing* involves sequential sensing by individual satellites within the same orbital plane, with handover used to maintain continuity. In contrast, *multi-orbital cooperative sensing* leverages simultaneous observations from multiple satellites across multiple orbits with overlapping

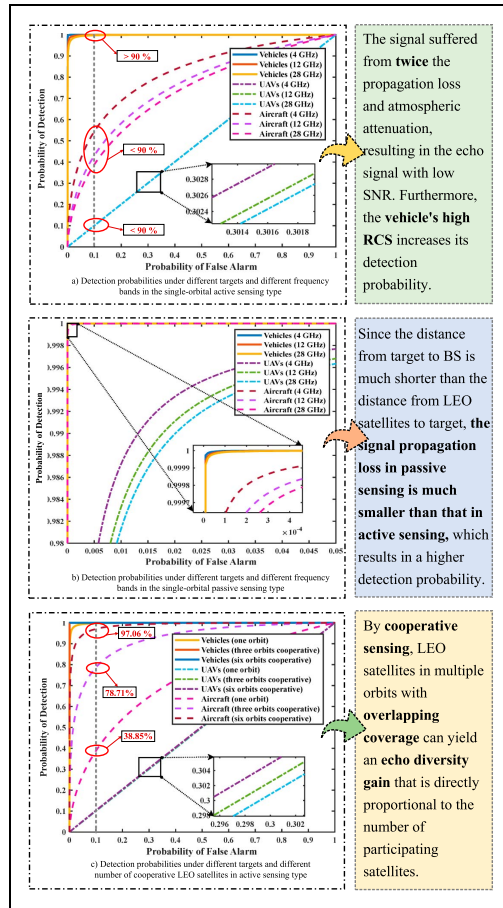


FIG. 2. The probabilities of detection for different types of networked sensing. Simulation parameters: the vehicle, UAV, and aircraft have RCS values of 50, 0.01, and 5 m^2 , and altitudes of 1.5, 200, and 10,000 m, respectively; the considered frequency bands are C (4 GHz), Ku (12 GHz), and Ka (28 GHz), reflecting typical satellite communication bands.

coverage, enabling cooperative sensing through sensing information fusion.

Each of the above sensing types offers unique benefits. As demonstrated in Fig. 2, both passive and hybrid sensing exhibit superior signal-to-noise ratio (SNR) performance compared to active sensing due to the shorter propagation paths of passive sensing compared to those of active sensing, which verifies that passive and hybrid sensing are particularly suitable for detecting targets with low RCS. Moreover, multi-orbital cooperative sensing significantly outperforms single-orbital sensing by enabling more reliable and accurate target observation through multi-satellite collaboration. It is important to note that overcoming the low-SNR echo challenge of active sensing through single-orbital and multi-orbital cooperative sensing is a critical open problem. Key research directions include the use of large onboard antenna arrays or MIMO systems to provide high beamforming gain, and the design of coded waveforms with high time-bandwidth products to achieve significant processing gain after pulse compression. The feasibility of these solutions, however, must always be balanced against the strict size, weight, power, and cost constraints of LEO satellite payloads.

B. USE CASES AND REQUIREMENTS

With the inherent advantages of LEO constellation networks in global coverage, low-latency communication,

and multi-node cooperative sensing, the ISAC-empowered LEO constellations play a pivotal role in several representative 6G application scenarios, as illustrated in Fig. 3.

- Low-Altitude Economy (LAE):** By exploiting its continuous sensing capability and robust space-to-ground LoS channels, the ISAC-empowered LEO constellation can support real-time three-dimensional (3D) trajectory reconstruction and dynamic airspace conflict alerts for UAV-based logistics services. These tasks demand high sensing accuracy, with range resolution below 5 m, range accuracy below 0.5 m, and refresh rate exceeding 10 Hz [10].
- Internet of Vehicles (IoV):** Through wide-area coverage and cooperative sensing, LEO constellations can enhance autonomous driving by enabling beyond-LoS obstacle detection, accurate target tracking, and high-fidelity road feature reconstruction. Required performance metrics include sub-meter positioning, velocity estimation accuracy better than 0.5 m/s, time synchronization accuracy better than 10 ns, and end-to-end latency smaller than 50 ms [10].
- Disaster Relief and Emergency Response:** The LEO constellation's wide-area visibility and persistent sensing make it ideal for supporting UAV swarms in post-disaster scenarios involving 3D mapping, autonomous path planning, and real-time search-and-rescue operations. These missions impose stringent requirements such as target tracking refresh rates above 10 Hz and closed-loop latency below 100 ms [10].
- Smart City:** ISAC-empowered LEO constellations can assist terrestrial infrastructure (e.g., BSs) in user behavior monitoring, real-time digital twin rendering, and urban intrusion detection. These services require meter-level sensing resolution and a detection probability exceeding 90% [10].

C. TECHNOLOGICAL FOUNDATION FOR NETWORKED SENSING

Waveform and payload design are two critical enablers for networked sensing in ISAC-empowered LEO constellations. This section discusses key challenges and potential solutions.

1) **Waveform Design:** As highlighted in Section I, traditional OFDM waveforms are inadequate for meeting the stringent sensing requirements of LEO constellations, particularly the requirements for extended maximum unambiguous range and velocity. Furthermore, waveform design needs to balance the trade-offs among PAPR, communication capacity, and sensing accuracy.

However, it is crucial to acknowledge OFDM's inherent sensitivity to the high Doppler shifts in LEO links. To address this fundamental challenge, the next-generation waveforms such as orthogonal time-frequency space (OTFS) and affine frequency-division multiplexing (AFDM) are being explored [11]. AFDM is particularly compelling for ISAC due to its intrinsic chirp-based structure, which is synergistic with radar sensing. While these represent future directions, improving OFDM's performance within the current framework remains a practical necessity.

Enhancing maximum unambiguous range and velocity: According to [12], the maximum unambiguous

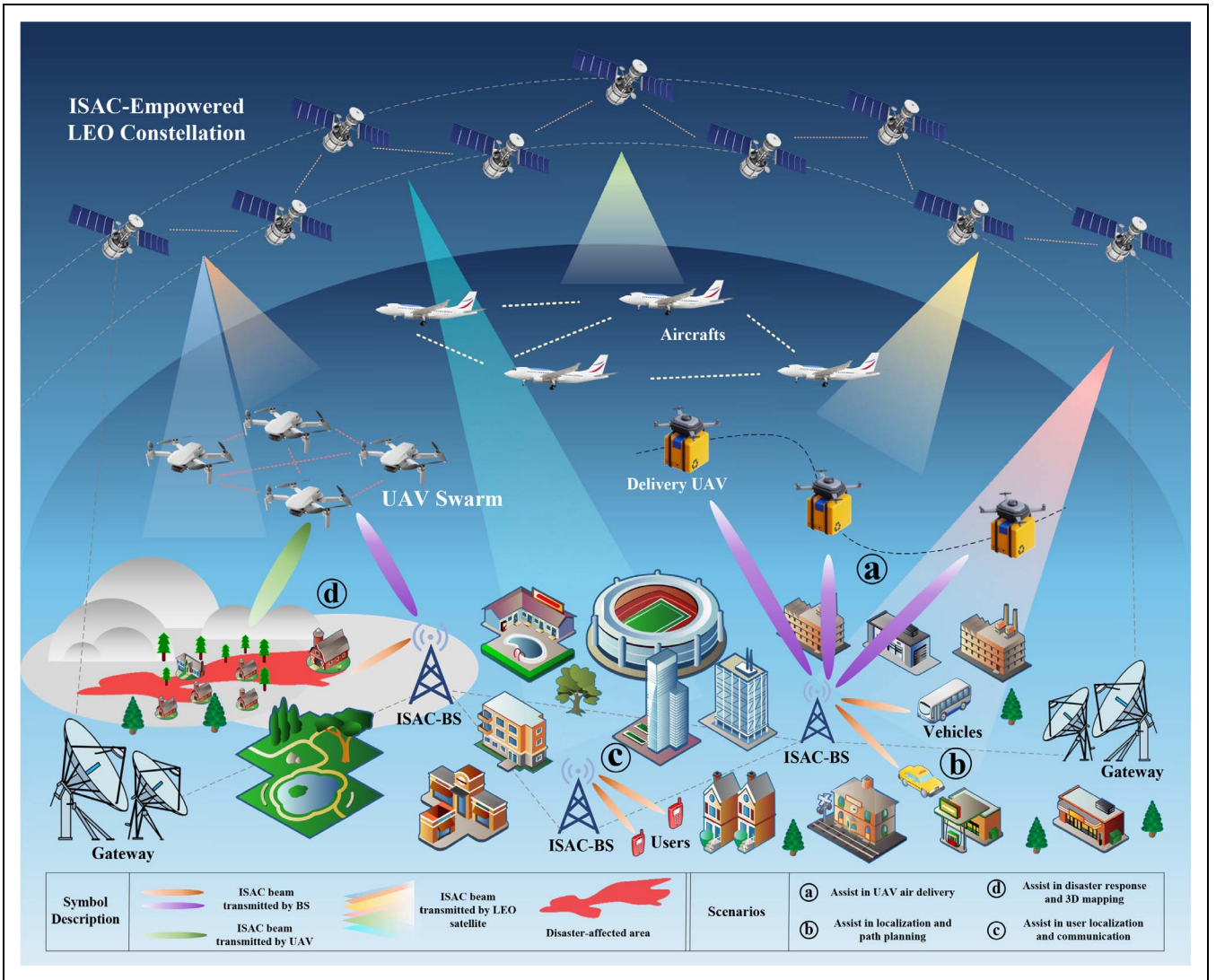


FIG. 3. Application scenarios of the ISAC-empowered LEO constellations.

range and velocity of an OFDM signal are

$$R_{\max} = \frac{cT_{\text{sym}}}{2}, \quad V_{\max} = \frac{c}{4f_c T_{\text{sym}}}, \quad (1)$$

where c denotes the speed of light, T_{sym} is the OFDM symbol duration, and f_c is the carrier frequency. According to 3GPP standard [2], standard OFDM waveforms cannot simultaneously satisfy the requirements of $R_{\max} > 500$ km and $V_{\max} > 6.9$ km/s.

To overcome this limitation within the existing 3GPP standard, we propose a carrier aggregation-based ISAC waveform [6]. This design aggregates three carefully selected component carriers (CCs) such that the greatest common divisor of their respective unambiguous ranges (or velocities) is minimized. Specifically, the proposed waveform includes one CC with 30 kHz subcarrier spacing in the 2 GHz band, and two CCs with 60 kHz subcarrier spacing in the 3.5 GHz band using regular and extended cyclic prefixes, respectively. This heterogeneity enables independent sensing on each CC.

By leveraging the Chinese remainder theorem (CRT), the range and velocity measurements across

the carriers are jointly reconstructed to significantly extend unambiguous sensing limits. Simulation results indicate that the proposed waveform achieves a maximum unambiguous range of 55,831 km and a velocity of 7.9 km/s, fully meeting LEO operational requirements. However, this approach demands high-performance radio frequency front-ends and powerful onboard baseband processors.

Balancing PAPR, communication reliability, and sensing accuracy: In LEO constellations, OFDM typically exhibits a PAPR of approximately 6 dB under quadrature phase shift keying (QPSK) modulation. However, such a high PAPR value leads to a minimum 40% reduction in link capacity [2]. The conventional clip-filter method reduces PAPR with significantly degraded communication and sensing performance [13].

To address this problem, we design a new ISAC waveform that achieves joint optimization by projecting the in-band frequency-domain components of the clipped signal onto a rectangular constraint set. Fig. 4 presents a performance comparison with the baseline cyclic prefix OFDM (named OFDM) and clip-filtered OFDM [13], demonstrating that the new

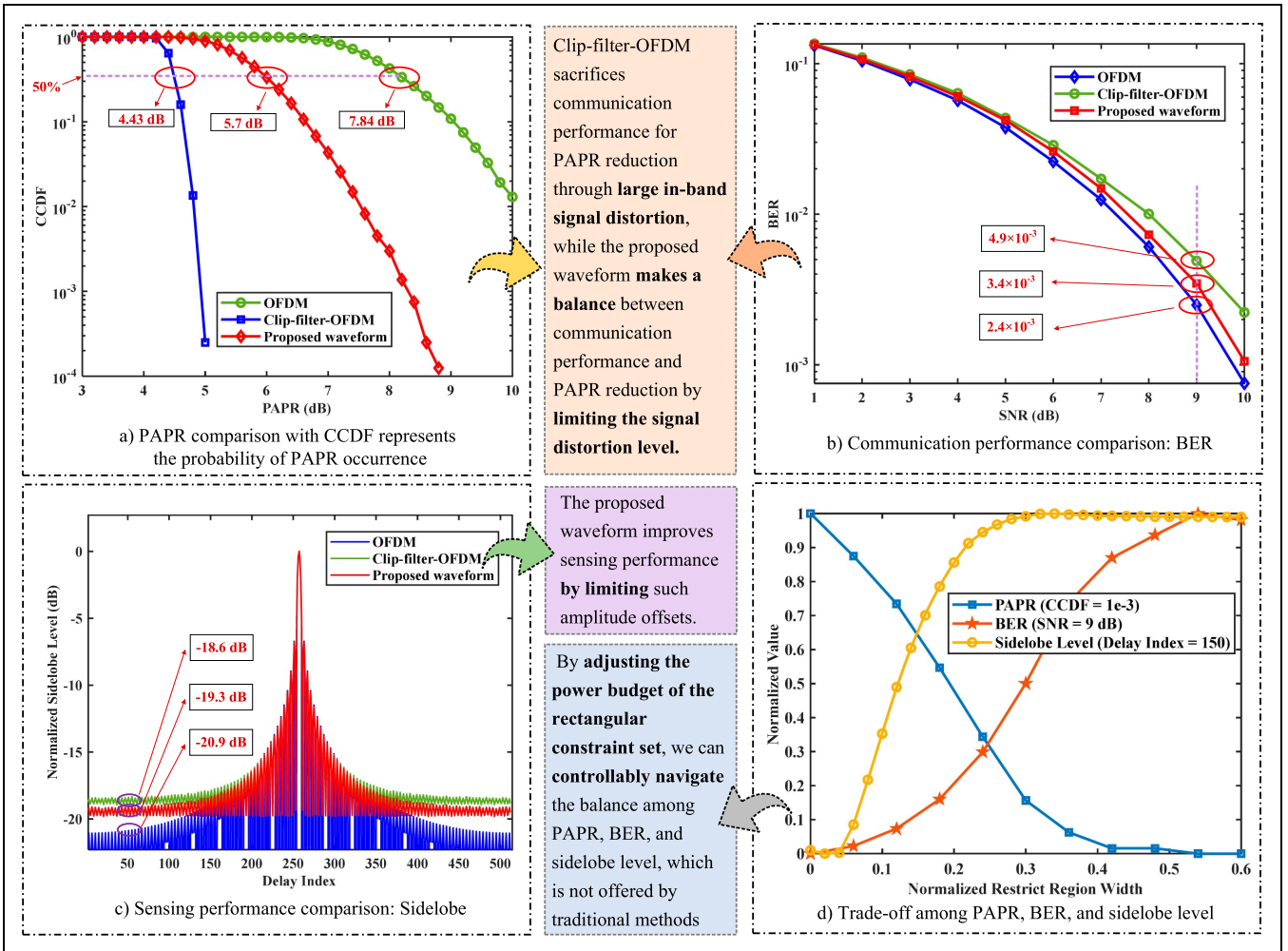


FIG. 4. The performance comparison about PAPR (complementary cumulative distribution function (CCDF)), communication (bit error rate (BER)), and sensing (sidelobe) with OFDM and clip-filter OFDM [13] as benchmarks. Simulation parameters are configured as follows: the modulation is QPSK, the number of subcarriers is 128, the number of symbols is 500, and clipping ratio of 1.5. This analysis is particularly relevant for ISAC in LEO constellations, as the satellites' strictly power-limited payloads demand low-PAPR waveforms for efficient power amplifier operation. The simulation demonstrates how the proposed method provides a controllable trade-off mechanism to balance these often conflicting requirements, which is a crucial capability for adaptive resource management in space.

ISAC waveform enables a favorable trade-off among PAPR, communication reliability, and sensing accuracy.

2) *Payload Enhancements:* To support real-time networked sensing and ISAC signal processing under orbital constraints, significant payload upgrades are required. These upgrades not only meet aerospace engineering standards but also accommodate the demands of ISAC.

Enhanced LEO satellite payload design: Computationally intensive signal processing and onboard artificial intelligence (AI) inference bring a computational burden to the traditional aerospace-grade processors under strict power budgets. To address this problem, we propose a heterogeneous computing platform combining the following components.

- A high-performance advanced RISC machines (ARM) central processing unit.
- A field-programmable gate array (FPGA) for triple-modular redundancy verification.
- A low-power AI accelerator capable of delivering over 100 tera operations per second (TOPS) under a 40W power envelope.

Moreover, conventional phased array systems incur beam switching delays in the order of milliseconds. To achieve microsecond-level responsiveness

required for tight coordination with terrestrial BSs, fully digital beamforming should replace mechanically scanned arrays. Additionally, traveling wave tube amplifiers, which operate at $\sim 30\%$ efficiency, should be replaced with Gallium Nitride-based solid-state power amplifiers, capable of exceeding 45% efficiency.

Enhanced BS design: To reduce latency and minimize satellite-ground hops, the traditional four-hop chain (satellite $\rightarrow$ gateway $\rightarrow$ core network $\rightarrow$ BS) should be replaced with direct satellite-to-BS links. To this end, the following design is essential.

- The antenna arrays of BSs need to be equipped with motorized elevation mechanisms, extending coverage to around $\pm 60^\circ$ for reliable down-link reception of LEO.
- Legacy baseband units (BBUs) need to be upgraded to NTN-compatible BBUs integrating wideband Doppler compensators and elastic timing controllers.

III. NETWORKED SENSING FOR TARGET SENSING

A prerequisite for realizing any multi-orbital cooperative sensing is the achievement of precise inter-satellite time-frequency synchronization. To

cooperatively process signals from different satellites, their local clocks and carrier frequencies must be aligned with extremely high precision. This is a significant challenge due to the factors such as independent clock drifts and orbital perturbations. Potential solutions involve leveraging global navigation satellite system signals for coarse common-time referencing, combined with inter-satellite links for fine-grained calibration. Developing robust and lightweight algorithms for this purpose remains a critical open research problem.

Beyond this foundational synchronization issue, several other significant challenges must be addressed to realize the full potential of multi-orbital cooperative sensing. The promise of this approach is substantial. As shown in Fig. 2, passive cooperative sensing offers superior capabilities for detecting challenging low-RCS targets (e.g., UAVs) for applications such as LAE and disaster management. Nevertheless, achieving the required high-accuracy performance consistently requires overcoming the following specific technical problem.

A. TIME-FREQUENCY SYNCHRONIZATION

Passive sensing benefits from a higher SNR due to shorter propagation paths than active sensing. However, its accuracy is ultimately constrained by time-frequency synchronization errors that arise from the separation between the transmitter and receiver. Although the LEO satellites are equipped with high-accuracy oscillators and pre-compensation techniques, the residual time offset (TO) and carrier frequency offset (CFO) typically remain at tens of nanoseconds and tens of hertz, respectively. The residual synchronization errors, primarily induced by oscillator imperfections such as phase noise and frequency drift, result in meter-level range estimation errors and meter-per-second-level velocity errors. For applications such as LAE and IoV, these errors are unacceptable for the following reasons. First, they corrupt the relative positioning accuracy required for safe separation and collision avoidance between closely-spaced agents (e.g., UAVs in a swarm or vehicles in a platoon). Second, such large uncertainties make it impossible to resolve and track multiple targets in dense environments.

A promising synchronization method is the *reference path method* [12], which exploits the strong space-to-ground line-of-sight (LoS) path between the transmitter and receiver as a reference. By cross-correlating the reference with the echo signal of the target, residual TO and CFO components can be effectively canceled, thereby enhancing synchronization accuracy. However, time-frequency synchronization without LoS links remains an open problem.

B. FUSION OF MULTIPLE ECHO SIGNALS

In addition to synchronization, the fusion of multiple echo signals from multiple LEO satellites is crucial to improve sensing accuracy. There are currently the following fusion methods.

- **Data-level fusion:** Data-level fusion methods fuse the sensing results of multiple satellites [4], which have low complexity but suffer from low estimation accuracy.
- **Symbol-level fusion:** Symbol-level fusion methods fuse the pre-processed, complex-domain sensing symbols of multiple satellites [12], trading higher complexity for a substantial boost in cooperative gain compared to data-level fusion.

However, these conventional approaches are not suitable for the passive sensing of LEO satellites, as they fail to reconcile the demands between computational complexity and performance gain. While data-level fusion has low complexity, it lacks the necessary noise resilience and processing gain in low-SNR environments. Symbol-level fusion provides the required performance gain. However, its high computational complexity is incompatible with the satellite's strict processing budget. To this end, we propose a *CNN-based fusion and sensing framework*, as illustrated in Fig. 5. This deep neural network exploits hierarchical connections to extract high-level features from received echo signals and adaptively learns the fusion weights of each node, thereby enabling both deep fusion and robust sensing performance.

To verify its effectiveness, we simulate the multi-orbital cooperative passive sensing scenario illustrated in Fig. 1, using the methods in [4] and [12] as benchmarks. Fig. 5 presents the root mean square error (RMSE) performance for location and absolute velocity estimation. The simulation results clearly demonstrate the superiority of the proposed CNN-based scheme and highlight the potential of deep learning in multi-satellite sensing information fusion. It is worth noting that the proposed network uses a powerful CNN architecture to achieve the superior performance demonstrated in the simulation results. We acknowledge the significant computational challenge of deploying such a model, which has 69.49 million parameters and a complexity of 2.36 GFLOPs, on resource-constrained LEO satellites. To address this challenge, our enhanced LEO satellite payload design, as detailed in Section II-C2, incorporates a dedicated low-power AI accelerator in the satellite payload. This accelerator, providing over 100 TOPS, is capable of executing the proposed model thousands of times per second, which is more than sufficient to meet real-time sensing requirements. This confirms the practical feasibility of our approach. This operational model follows a practical offline/online paradigm: the computationally intensive training of the CNN is performed offline on the ground, while the satellite only needs to execute fast, low-latency online inference using the pre-trained model. Future work could also explore model compression techniques to further reduce the computational footprint.

IV. NETWORKED SENSING FOR TARGET TRACKING

Target tracking is essential to realize the continuous monitoring of the target. However, the rapid orbital mobility and dynamic topology of LEO constellations introduce challenges for target tracking, which are revealed in the following subsections.

A. TARGET TRACKING ALGORITHMS

Due to the high orbital velocity of the LEO satellite, the LEO satellite detects a target within a limited observation window, requiring highly accurate and computationally efficient target tracking algorithms. Moreover, the fast relative motion between the satellite and target causes rapid and nonlinear variations in range and Doppler frequency shift, rendering traditional algorithms such as the Kalman filter invalid.

To address these problems, the *physics-informed neural network* (PINN) can be utilized, which applies a model and data-driven method for target tracking. This method learns complex dynamics from observed data while adhering to known physical laws, ensuring

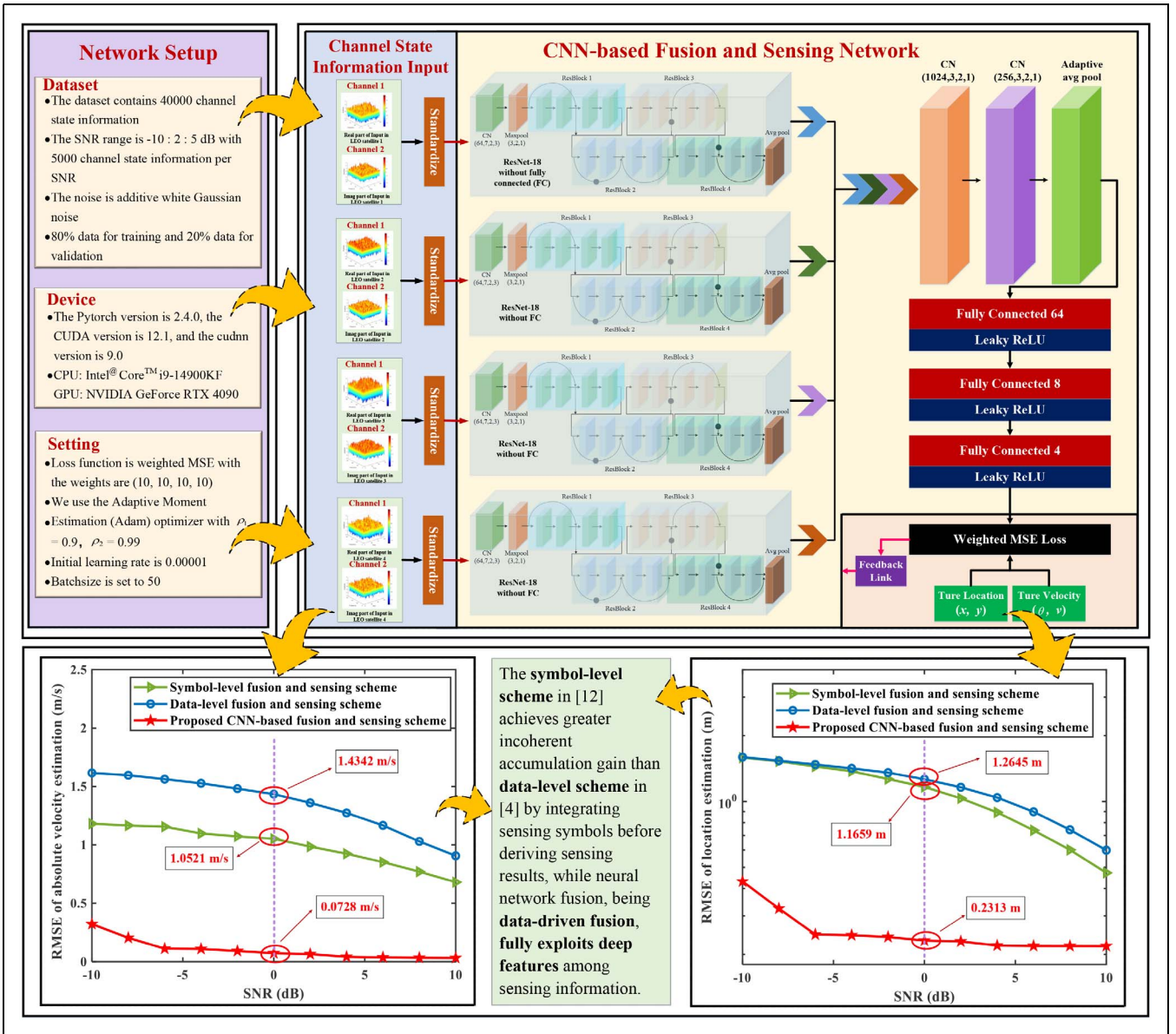


FIG. 5. The simulation of RMSEs among the proposed scheme, the scheme in [4], and the scheme in [12]. The simulation models a representative scenario with 4 LEO satellites at a 500 km altitude, a common setup in dense constellations. Following NTN standards [2], the carrier frequency is 4 GHz (C-band) and the subcarrier spacing is 60 kHz, with 128 subcarriers and symbols. Importantly, the chosen SNR range of -10 to 5 dB reflects the challenging yet realistic link budget resulting from the significant free-space path loss (~ 150 dB) at this altitude and weak echoes from targets.

both high accuracy of target tracking and physical plausibility. The primary advantage of PINN is the combination of data-driven machine learning and physical laws. On one hand, it learns directly from the satellite's observations, allowing it to capture complex target movements that traditional models might miss. On the other hand, it applies known physical laws into the learning model, such as the motion equations of the satellite and target, ensuring that the predictions of the neural network follow the physical laws. As a result, the PINN can generate highly accurate and realistic paths of targets even with the limited and noisy data of LEO satellite observations.

B. HANDOVER NODE SELECTION

Even with high-accuracy target tracking algorithms, achieving seamless target tracking requires optimal node handovers across the LEO constellations. To ensure over 99.9% tracking continuity, the system must perform real-time decision-making under a

rapidly changing constellation topology. Traditional handover decisions are typically formulated as optimization problems involving multiple criteria such as sensing continuity, link quality, target observation duration, and latency. However, in the dynamic environment of LEO constellations, where both the candidate handover node set and evaluation metrics change rapidly, the classical optimization approaches are inadequate due to their lack of timeliness and adaptability.

To address this problem, we propose a *distributed reinforcement learning* (RL) framework [14], wherein each satellite acts as an autonomous agent. Through extensive simulation-based training, each agent learns a policy network that maps local observations (e.g., relative location, link quality) directly to optimal handover actions for real-time online decision-making, with the objective of maximizing long-term tracking continuity in multiple handovers. Nevertheless, simulation-trained models struggle to generalize in

the presence of real-world phenomena such as atmospheric attenuation, radio interference, and user-specific sensing requirements. To overcome this problem, we introduce a two-stage strategy combining *digital twin* and *transfer learning* techniques. First, a high-fidelity digital twin environment is constructed to accurately simulate satellite behavior, enabling large-scale offline pre-training and validation. Then, transfer learning is employed to fine-tune the learned policy using limited real-world interaction data, allowing the model to rapidly adapt to environmental conditions that are not captured in simulation. This approach enhances both robustness and real-time applicability in actual deployment scenarios.

V. FUTURE RESEARCH TRENDS

Several open research challenges and potential solutions for ISAC-empowered LEO constellations are discussed in this section.

A. AI-BASED COOPERATIVE ISAC WAVEFORM DESIGN

In ISAC-empowered LEO constellations, cooperative waveform design across multiple satellites is essential to achieve network-wide sensing and communication gain. The cooperative waveform design needs to jointly optimize multi-dimensional resources across distributed nodes, while adhering to stringent performance constraints.

The AI technology offers a promising solution to this high-dimensional, tightly coupled optimization problem. Specifically, a graph neural network (GNN) can model the dynamic topology of the LEO constellation, capturing the inter-satellite dependencies and task correlations. Based on the learned cooperation strategy, a CNN can serve as a waveform generator for each satellite, generating baseband signals tailored to current network conditions.

This GNN-CNN hybrid framework enables an end-to-end mapping from network states to cooperative waveform design, effectively balancing the trade-offs between communication and sensing. However, the dynamic nature of the LEO constellation demands highly efficient and adaptive AI models. Crucially, while the complex training of such models would occur on the ground, future research must focus on designing the resulting waveform generators to be lightweight enough for real-time onboard inference. Moreover, the large scale of the LEO constellation complicates both training and deployment procedures of AI models. Ensuring generalization to new scenarios and improving model interpretability are also crucial for trustworthy AI model deployment.

B. MULTI-MODAL INTELLIGENT SENSING

The sensing capabilities of the ISAC-empowered LEO constellation, though powerful in terms of spatial-temporal coverage and continuity, are limited to radio-frequency modalities and cannot capture semantic information. To overcome this limitation, it is promising to integrate multi-modal sensors, such as remote sensing cameras, LiDAR, and infrared imagers, into the LEO satellite, forming a LEO constellation with multi-modal intelligent sensing. To address the inherent challenges in multi-modal information fusion, AI-driven feature-level, decision-level, and task-level fusion methods can be utilized.

However, real-time fusion under power and computation constraints, accurate spatiotemporal synchronization across multiple modalities in dynamic environments, and the acquisition of large-scale

annotated datasets are still challenging in multi-modal intelligent sensing of LEO constellations.

C. SPACE SITUATIONAL AWARENESS

With the increasing number of space activities, the crowded orbital environment has increased the collision risk of in-orbit satellites. To effectively safeguard these assets and prevent potential collisions, there is a pressing requirement for a comprehensive and precise awareness of the space environment. To meet this demand, the sensing scope of the ISAC-empowered LEO constellation should extend beyond terrestrial and aerial targets to encompass the broader domain of space situational awareness (SSA), including cooperative satellites, non-cooperative probes, and space debris.

To enable SSA, an ISAC-empowered LEO constellation can be configured as a cooperative sensing network that supports joint detection, target trajectory prediction, and collision risk assessment. Nevertheless, the tracking task of space targets introduces unique difficulties due to the following factors. i) extremely low RCS, long distance, and high velocities of space targets; ii) severe channel impairments caused by ionospheric scintillation and cosmic noise. Therefore, robust signal processing and target tracking algorithms are essential for effective SSA.

D. ENERGY-EFFICIENT RESOURCE SCHEDULING

LEO satellites are inherently energy-constrained, and traditional “always-on” sensing strategies are unsustainable in large-scale deployments. ISAC significantly reduces satellite lifespan and increases operational costs. Hence, energy-efficient resource scheduling mechanisms are required to dynamically activate the minimal set of satellites to perform sensing tasks, while maintaining global sensing coverage and smooth task handover.

Multi-agent reinforcement learning (MARL)-based methods that incorporate orbital dynamics and energy consumption models can autonomously manage sensing and communication resources in the LEO constellation by regarding each satellite as an agent [15]. However, realizing this vision requires the joint optimization of beam alignment, waveform design, and computation offloading adaptively guided by real-time energy budgets and mission priorities. Although MARL-based methods have been validated to be effective on ground simulation platforms, their effectiveness when transferred to space constellations remains challenging. A practical approach to overcoming this involves complex offline training of the MARL agents on the ground, allowing the satellites to execute only the resulting lightweight policy networks for fast, online decision-making.

VI. CONCLUSION

ISAC-empowered LEO constellation has emerged as a prominent research field. This article provides a comprehensive overview of the challenges, enabling technologies, and potential solutions for the networked sensing of the ISAC-empowered LEO constellation. We first reviewed the fundamental concepts, sensing types, application scenarios, and payload support for networked sensing of the ISAC-empowered LEO constellation. Subsequently, we explored technical challenges related to waveform design, high-accuracy target sensing, and real-time target tracking, along with the solutions. To demonstrate the feasibility and effectiveness of

these techniques, representative simulation results were presented and discussed. In addition, we highlighted several future research trends, including AI-based cooperative ISAC waveform design, multi-modal intelligent sensing, space situational awareness, and energy-efficient resource scheduling. This article may inspire the research interest in the networked sensing of ISAC-empowered LEO constellation.

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